One could stubbornly maintain that it's not good enough to show that people should act "as if" something is true; it needs to actually be true. That's missing the point a little bit. Many-Worlds quantum mechanics presents us with a dramatically different view of reality from an ordinary one-world view with truly random events. It's unsurprising that some of our most natural-seeming notions are going to have to change along with it. If we lived in the world of textbook quantum mechanics, where wave-function collapse was truly random and obeyed the Born rule, it would be rational to calculate our expected utility in a certain way. Deutsch and Wallace have shown that if we live in a deterministic Many-Worlds universe, it is rational to calculate our expected utility in exactly the same way. From this perspective, that's what it means to talk about probability: the probabilities of different events actually occurring are equivalent to the weighting we give those events when we calculate our expected utility. We should act exactly as if the probabilities we're calculating apply to a single chancy universe; but they are still real probabilities, even though the universe is a little richer than that.

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* There are more sophisticated arguments that such a rule follows from very weak assumptions. Wojciech Zurek has proposed a way of deriving such a principle, and Charles Sebens and I put forward an independent argument. We showed that this rule can be derived by insisting that the probabilities you assign for doing an experiment in your lab should be independent of the quantum state elsewhere in the universe.